

1 Background

Finite distributed lags

$y = \beta_1 + \sum_{i=0}^q \delta_i x_{t-i} + e_t$ (order q).

Impact multiplier: δ_0 , dynamic multiplier: $\delta_i, i \geq 1$, long run multiplier: $\theta_0 = \sum_{i=1}^q \delta_i$. Reparametrise s.t. $\delta_0 = \theta_0 - \sum_{i=1}^q \delta_i$, so $y = \beta_1 + \theta_0 x_t + \sum_{i=1}^q \delta_i (x_{t-i} - x_t)$. Obtain $\hat{\theta}_0$ and $se(\hat{\theta}_0)$ by regression.

Lagged dependent variables

AR(1): $y_t = \beta_1 + \beta_2 y_{t-1} + u_t$.

When $|\beta_2| < 1$, equiv. to $y_t = \beta_1/(1 - \beta_2) + e_t + \beta_2 e_{t-1} + \dots = \beta_1/(1 - \beta_2) + \sum_{i=0}^T \beta_2^i e_{t-i}$.

OLS estimator: $\hat{\beta} = (X^T X)^{-1} X^T y = \frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n (x_i)^2} = \frac{\sum_{i=1}^n (x_i - \bar{x}) y_i}{\sum_{i=1}^n (x_i - \bar{x})^2}$.

OLS estimators biased; need large sample analysis, which is harder w/ time corr.

Gauss-Markov: $rank(X) = k$, $E[e|X] = 0$, $Var(e|X) = \sigma^2 I_T$, then OLS est. is BLUE. $E[e_t|X] = 0 \forall t$ means y_{t-1} can't be a regressor and y_{t-1} can't impact x_t .

If y_{t-1} affects x_t , then $Cov(x_t, e_{t-1}) \neq 0$.

If y_{t-1} is a reg, like in AR(1), $Cov(e_t, y_t) = Cov(e_t, \beta_1/(1 - \beta_2) + \sum_{i=0}^T \beta_2^i e_{t-i}) = \sigma^2 \neq 0$.

$Var(e|X) = \sigma^2 I_T \Rightarrow Var(e_t|X) = \sigma^2, \forall t$ (homo.), $Cov(e_t, e_s|X) = 0 \forall t \neq s$ (no serial corr.). Homot: $Var(e_t)$ indep. of X and const. over time. No serial corr.: reg. errors white noise. y_t, x_t can be serially corr. $Var(\hat{\beta}|X) = \sigma^2 (X^T X)^{-1}$, and an estimate of σ^2 is $\hat{\sigma}^2 = 1/(T - k) \sum_{t=1}^T \hat{e}_t^2$.

$se(\hat{\beta}_j) = \hat{\sigma} / \sqrt{SST_j(1 - R_j^2)}$ where R_j^2 is the R^2 from reg of x_j on remaining regs and $SST_j = \sum_{t=1}^T (x_{jt} - \bar{x}_j)^2$.

Above formulas don't apply w/ hetero. and serial corr.

2 Time series analysis

Cov-stationary: $\{y_t\}$ is cov/weakly stnry if $E(y_t) = \mu$ and $\gamma_{k,t} = Cov(y_t, y_{t-k}) = E[(y_t - \mu)(y_{t-k} - \mu)] = \gamma_k$ indep. of t .

γ_k : autocov. fn, $\rho_k = \gamma_k/\gamma_0$: autocorr. fn. For a stationary process, $\gamma_0 = \gamma_{-k}$

$Cov(X, Y) = E[XY] - E[X]E[Y]$.

$\{y_t\}$ white noise if $\mu = 0$ and $\gamma_k = 0 \forall k \neq 0$.

Indep. over time, then indep. white noise.

$\{y_t\}$ is strictly stnry if joint distr. $(y_t, y_{t-1}, \dots, y_{t-k})$ indep. of $t \forall k$. Strict stnry $\Rightarrow$ cov-stnry, but opp. is not true, since higher order moments can dep. on time.

Gaussian process characterised by μ and cov matrix, so strict iff weak stnry.

Any transformation of strict stnry process is strictly stnry.

Weak dependence: $\gamma_k \rightarrow 0$ as $k \rightarrow \infty$. For LLN and CLT, $\gamma_k \rightarrow 0$ s.t. $\sum_{k=0}^{\infty} |\gamma_k| < \infty$.

MA and AR Models

MA(1): $y_t = \mu + \varepsilon + \theta_1 \varepsilon_{t-1}$, ε_t WN. $E(y_t) = \mu$, $\gamma_0 = (1 + \theta_1^2)\sigma^2$, $\gamma_1 = \theta_1\sigma^2$, $\gamma_k = 0 \forall k > 1$. MA(1) stnry and weak dep $\forall \mu, \theta$. ρ_1 attains max at $\theta_1 = 1$ and min at $\theta_1 = -1$ by FOC. $\forall \rho_1$, 2 vals of θ_1 that produce it, θ_1 and $1/\theta_1$.

MA(q): $y_t = \mu + \varepsilon + \theta_1 \varepsilon_{t-1} + \dots + \theta_q \varepsilon_{t-q}$, ε_t WN. $\gamma_k = 0$ for $k > q$ and stnry.

Causal: Can express y_t as a fn of ε_t and past ε 's. MA(q) causal w/ no restrictions.

MA(∞): $y_t = \mu + \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}$, ε_t WN and $\{\psi_j\}$ sequence s.t. $\psi_0 = 1$. If $\sum_{j=0}^{\infty} |\psi_j| < \infty$, y_t cov-stnry and weak dep. For cov-stnry, $E(y_t) = \mu$ and $\gamma_k = \sigma^2 \sum_{j=0}^{\infty} \psi_j \psi_{j+k}$ for $k \geq 0$. For weak dep., $\sum_{k=1}^{\infty} |\gamma_k| < \infty$.

AR(1): $y_t = c + \phi_1 y_{t-1} + \varepsilon_t$, ε_t WN.

Stnry iff $|\phi_1| < 1$, since $y_t \sim MA(\infty)$. If $|\phi_1| \geq 1$, no causal stnry soln. If $\phi_1 = 1, c = 0$, random walk w/ $y_t = y_{t-1} + \varepsilon_t = y_0 + \sum_{i=1}^t \varepsilon_i$, not stnry bc $Var(y_t) = t\sigma^2$.

If $|\phi_1| < 1$, $y_t = c(1 - \phi_1^m)/(1 - \phi_1) + \phi_1^m y_{t-m} + \sum_{i=0}^{m-1} \phi_1^i \varepsilon_{t-i}$. As $m \rightarrow \infty$, $y_t = \frac{c}{1 - \phi_1} + \sum_{i=0}^{\infty} \phi_1^i \varepsilon_{t-i}$. $|\phi_1| < \infty$.

Lag op: $L^j x_t = x_{t-j}$, $L^j a = a$ if a const.

$1/(1 - \phi_1 L) = \sum_{j=0}^{\infty} \phi_1^j L^j$.

For AR(1), $(1 - \phi_1 L)y_t = c + \varepsilon_t$. Stable if $|\phi_1| < 1$, $\Leftrightarrow \phi(z) = 1 - \phi_1 z = 0$ when $|z| = |1/\phi_1| > 1$. For stable AR(1), $IRF_h = \phi_1^h$, $LRM = 1/(1 - \phi_1)$, $\gamma_k = \phi_1^k \sigma^2/(1 - \phi_1^2) \forall k \geq 0$, and $\rho_k = \phi_1^k$, where $\sigma^2 = Var(\varepsilon_t)$.

$E(y_t) = c + \phi_1 E(y_{t-1}) + E(\varepsilon_t) \Rightarrow (1 - \phi_1)E(y_t) = c \Rightarrow E(y_t) = c/(1 - \phi_1)$.

Yule-Walker eqn: $\gamma_k = \phi_1 \gamma_{k-1}$.

So $\sum_{k=1}^{\infty} |\gamma_k| = \gamma_0 \sum_{k=0}^{\infty} |\phi_1|^k = \gamma_0/(1 - |\phi_1|) < \infty$, and process is weak. depndt.

AR(p): $y_t = c + \sum_{i=1}^p \phi_i y_{t-i} + \varepsilon_t$, or in lag op. terms, $(1 - \sum_{i=1}^p \phi_i L^i)y_t = c + \varepsilon_t$.

Stnry cond is p roots of $\phi(z) = 1 - \sum_{i=1}^p \phi_i z^i = 0$ outside unit circle ($|z| > 1$).

$p = 2$, $\phi_1 + \phi_2 < 1$ necssry but insuffnt.

W/ stability, AR(2) $\sim$ MA(∞), where $y_t = \frac{c}{1 - \phi_1 - \phi_2} + \sum_{i=0}^{\infty} \psi_i \varepsilon_{t-i}$, causal and stnry.

ψ_i 's fn of ϕ_1, ϕ_2 st $\psi_0 = 1$, $\sum_{i=0}^{\infty} |\psi_i| < \infty$. To get MA coeffs, $\phi(L)y_t = \varepsilon_t \Rightarrow y_t = \phi(L)^{-1} \varepsilon_t \Rightarrow \psi(L) = \phi(L)^{-1}$. Solve $\phi(L)\psi(L) = 1$ to get $\psi(L)$ terms.

W/ MA representation, $E(y_t) = \frac{c}{1 - \phi_1 - \phi_2}$.

Yule-Walker: $\gamma_k = \phi_1 \gamma_{k-1} + \phi_2 \gamma_{k-2}$.

Invertible: x_t depends on $y_t, y_{t-1}, \dots$ (can be written as AR(∞)). AR always, MA invertible if coeffs restricted.

For MA(1), $y_t = \varepsilon_t + \theta \varepsilon_{t-1}$. If $|\theta| < 1$, $y_t = \varepsilon_t + \sum_{i=1}^{\infty} \theta^i y_{t-i}$, an AR(∞).

$y_t = \varepsilon_t + \theta \varepsilon_{t-1} = \theta(L)\varepsilon_t$, where $\theta(L) = 1 + \theta L$. $\theta(L)^{-1} = \sum_{i=0}^{\infty} (-\theta L)^i$, so invert. cond. is $\theta(z) = 1 - \sum_{i=1}^q \theta_i z^i = 0$ root s.t. $|z| > 1$.

We get $\theta(L)^{-1} y_t = (1 - \theta L + \theta^2 L^2 - \dots) y_t = \sum_{j=0}^{\infty} (-\theta)^j y_{t-j} = \varepsilon_t$. ε_t causal fn of y_t .

AR mdl's easily estimated by OLS, so turning MA into AR helps us estimate θ .

Probability

plim $\hat{\theta}_n = \theta$: $\forall \varepsilon > 0$, $\lim P(|\hat{\theta}_n - \theta| < \varepsilon) = 1$. LLN: $\text{plim}(1/n \sum_{i=1}^n X_i) = E(X_i) = \mu$.

$\forall g \in C^1$, $\text{plimg}(\hat{\theta}_n) = g(\text{plim} \hat{\theta}_n)$.

$\text{plim} 1/n \sum_{i=1}^n (X_i - \bar{X})^2 = Var(X_i)$, and $\text{plim} 1/n \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}) = Cov(X_i, Y_i)$.

Since LLN holds for stnry and weak dep. data, $\text{plim} \hat{\gamma}_k = \text{plim} \frac{1}{T} \sum_{t=1+k}^T (y_t - \bar{y})(y_{t-k} - \bar{y}) = \gamma_k$.

$\lim E[(y_t - \bar{y})^2] = 0 \Rightarrow$ conv in prob. So $E(\bar{y}) = \mu$ and $Var(\bar{y}) \rightarrow 0 \Rightarrow \text{plim} \hat{\gamma}_k = \gamma_k$.

Weak exog: $E(x_t | e_t) = 0, \forall t$, so $Cov(x_{jt}, e_t) = 0 \forall j, t$. But $Cov(x_{t+1}, e_t)$ not necessarily = 0.

Consistent: $\text{plim} \hat{\beta} = \beta$. OLS $\hat{\beta}$ is consistent w/ $Cov(x_t, e_t) = 0$, by weak exog.

Conv. in distribution: Z_n, Z rvs, $F_n(x) = P(Z_n \leq x)$, $F(x) = P(Z \leq x)$. $Z_n \xrightarrow{d} Z$ if $\lim F_n(x) = F(x) \forall x$ where $F(x)$ cont.

CLT: $\{Z_i\}_{i=1}^n$ iid rvs, $E(Z_i) = \mu < \infty$, $Var(Z_i) = \sigma^2 < \infty$, then $\frac{\bar{Z}_n - E(\bar{Z}_n)}{\sqrt{Var(\bar{Z}_n)}} \xrightarrow{d} N(0, 1)$.

Equiv, $\sqrt{n} \frac{\bar{Z}_n - \mu}{\sigma} \xrightarrow{d} N(0, 1)$.

$Z_n \xrightarrow{P} Z \Rightarrow Z_n \xrightarrow{d} Z$.

Asymp. dist of $\hat{\beta}_2$: $\sqrt{n}(\hat{\beta}_2 - \beta_2) = \frac{1/\sqrt{n} \sum_{i=1}^n (x_i - \bar{x}) e_i}{1/n \sum_{i=1}^n (x_i - \bar{x})^2} \xrightarrow{P} \frac{Var(x_i)}{1/n \sum_{i=1}^n (x_i - \bar{x})^2}$. Num: $1/\sqrt{n} \sum_{i=1}^n (x_i - \bar{x}) e_i = 1/\sqrt{n} \sum_{i=1}^n (x_i - \mu) + (\mu - \bar{x})$.

$\xrightarrow{P} 1/\sqrt{n} \sum_{i=1}^n e_i \xrightarrow{P} 1/\sqrt{n} \sum_{i=1}^n (x_i - \mu) e_i \xrightarrow{d} N(0, Var((x_i - \mu) e_i))$. $\hat{\beta}_2 \xrightarrow{d} N(0, V)$, $V = \frac{Var((x_i - \mu) e_i)}{(Var(x_i))^2}$. If homo, $V = \frac{\sigma^2}{Var(x_i)}$.

Conv. rate: $V = Var(\sqrt{n} \hat{\beta}_2)$, so $Var(\hat{\beta}_2) = V/n$. $\hat{\beta}_2$ is root- n consistent.

Time series case: $\sqrt{T}(\hat{\beta}_2 - \beta_2) = \frac{1/\sqrt{T} \sum_{t=1}^T (x_t - \bar{x}) e_t}{1/T \sum_{t=1}^T (x_t - \bar{x})^2} \xrightarrow{P} \frac{A}{1/T \sum_{t=1}^T (x_t - \bar{x})^2}$.

$A = (Var(x_t))$, const. $1/\sqrt{T} \sum_{t=1}^T (x_t - \bar{x}) e_t \xrightarrow{d} N(0, B)$. $Var(\sqrt{T} \hat{\beta}_2) = A^{-1} B A^{-1}$. $\hat{\mu} | e_t \xrightarrow{d} N(0, B)$. $Var(\sqrt{T} \hat{\beta}_2) = A^{-1} B A^{-1}$. $\hat{\mu} = 1/T \sum_{t=1}^T (x_t - \bar{x})^2$.

Hypo test: $H_0: \beta_2 = 0$ vs $H_1: \beta_2 \neq 0$. $\sqrt{T}(\hat{\beta}_2 - \beta_2) \xrightarrow{d} N(0, A^{-1} B A^{-1})$. Under H_0 , $\sqrt{T} \hat{\beta}_2 / \sqrt{V} \xrightarrow{d} N(0, 1)$. If $\hat{V} \xrightarrow{P} V$, $t = \sqrt{T} \hat{\beta}_2 / \sqrt{\hat{V}} \xrightarrow{d} N(0, 1)$, since $\sqrt{V} / \sqrt{\hat{V}} \xrightarrow{P} 1$. α is $P(\text{reject } H_0 | H_0 \text{ true})$. Reject if $|t| > 1.96$ for $\alpha = 0.05$.

Variance Estimation
Long run var: $B = Var(1/\sqrt{T} \sum_{t=1}^T z_t)$, where $z_t = (x_t - \mu) e_t$. For iid data, $B = Var((x_t - \mu) e_t)$, where $\mu = E(x_t)$, can estimate w/ White var est.

Para: Know cov. struct of z_t , est z_t w/ this.

Nonparametric: use HAC estimator

$\bar{z} = 1/T \sum_{t=1}^T z_t$, w/ $E(z_t) = 0$.

$Var(\bar{z}) = E((1/T \sum_{t=1}^T z_t)^2) = 1/T^2 E(\sum_{t=1}^T \sum_{s=1}^T z_t z_s) = \sum_{t=1}^T z_t^2 + 2 \sum_{k=1}^{T-1} \sum_{t=1}^{T-k} z_t z_{t+k}$. So, $T Var(\bar{z}) = Var(\sqrt{T} \bar{z}) = 1/T \sum_{t=1}^T E(z_t^2) + 2 \sum_{k=1}^{T-1} 1/T \sum_{t=1+k}^T E(z_t z_{t-k})$. W/ stnry, $Var(\sqrt{T} \bar{z}) = \gamma_0 + 2 \sum_{k=1}^{T-1} (1 - k/T) \gamma_k$. $B = \lim Var(\sqrt{T} \bar{z}) = \gamma_0 + 2 \sum_{k=1}^{\infty} \gamma_k$. B for AR(1): $z_t = \rho z_{t-1} + u_t$, $u_t \sim WN(0, \sigma^2)$, $|\rho| < 1$. $\gamma_0 = \sigma^2/(1 - \rho^2)$, $\gamma_k = \rho^k \gamma_0$. $B = \frac{\sigma^2}{1 - \rho^2} + 2 \sum_{k=1}^{\infty} \rho^k \frac{\sigma^2}{1 - \rho^2} = \frac{\sigma^2}{(1 - \rho^2)^2}$.

$\hat{B}_{AR(1)} = \frac{\hat{\sigma}^2}{(1 - \hat{\rho})^2}$, parametric estimator.

Nonpara: If $\{z_t\}$ stnry with $\sum_{k=1}^{\infty} \gamma_k < \infty$ (weak dep), $B = \lim Var(\sqrt{T} \bar{z}) = \gamma_0 + 2 \sum_{k=1}^{\infty} \gamma_k$, where $\gamma_k = E(z_t z_{t-k})$ if $E(z_t) = 0$. For $k \leq T - 1$, $\hat{\gamma}_k = 1/T \sum_{t=1+k}^T z_t z_{t-k}$.

Naive, truncate: $\hat{B}_{trunc} = \hat{\gamma}_0 + 2 \sum_{k=1}^M \hat{\gamma}_k$ for $M < T$. M is bandwidth/lag trunc param. If $\gamma_k = 0 \forall |k| > M$, then consis. Otherwise, $M \rightarrow \infty$ since $Bias(\hat{B}) \approx 1/M$. But $Var(\hat{B}) \propto M/T$ blows up.

If $M = 1$, then $\hat{B}_{trunc} = \hat{\gamma}_0 + 2 \hat{\gamma}_1$, and $\hat{B}_{trunc} < 0$ if $\hat{\gamma}_1 < -1/2 \hat{\gamma}_0$. Can add weights $k_T(j)$ s.t. estimator is +ve, leading to HAC kernel estimators.

HAC estimator: $\hat{B} = \hat{\gamma}_0 + 2 \sum_{j=1}^{T-1} k(\frac{j}{M}) \hat{\gamma}_j$, where $k(x)$ is kernel fn, M is bandwidth param, and $\hat{\gamma}_j = 1/T \sum_{t=1+j}^T (z_t - \bar{z})(z_{t-j} - \bar{z})$.

Trunc. ker: $k(x) = 1$ if $|x| \leq 1, 0$ if $|x| > 1$.

Bartlett (NW) ker $k(x) = 1 - |x|$, if $|x| \leq 1, 0$, if $|x| > 1$.

Both ker s.t. when $|j/M| > 1$, $k(j/M) = 0$, so M also called truncation lag.

Quad spectr and Bartlett ker give +ve var.

Choose M to minimise MSE of $\hat{B}$, so $\min E((\hat{B} - B)^2) = Bias(\hat{B})^2 + Var(\hat{B})$.

For NW ker, $Bias(\hat{B}) = c_1/M$, so $Bias(\hat{B})^2 = c_1^2/M^2$. $Var(\hat{B}) = c_2(M/T)$. So $MSE(\hat{B}) = c_1^2(1/M^2) + c_2(M/T)$. W/ FOC, $M = d(T^{1/3})$, d const dpndt on serial corr struct of z_t .

Can est d based on AR(1) mdl for z_t w/ ρ . For NW ker: $d = 0.75 T^{1/3}$ when $\rho = 0.5$.

HAC var estim: $\widehat{Var}(\sqrt{T} \hat{\beta}_2) = \hat{A}^{-1} \hat{B} \hat{A}^{-1}$, $\hat{A}$ is samp. var of x_t , $\hat{B} = \hat{\gamma}_0 + 2 \sum_{j=1}^{T-1} k(j/M) \hat{\gamma}_j$, where $\gamma_j = 1/T \sum_{t=1+j}^T (x_t - \bar{x})(x_{t-j} - \bar{x}) \hat{e}_t \hat{e}_{t-j}$.

Special case: White var estimator, $\hat{B} = \hat{\gamma}_0$.

Martingales

No serial corr, but not indep over time.

z_t vec and $x_t \in z_t$. $\{x_t\}$ martingale wrt $\{z_t\}$ if $E(x_t | \mathcal{F}^{t-1}) = z_{t-1}, z_{t-2}, \dots = x_{t-1} \forall t$.

If $z_t = x_t$, x_t martingale w.r.t. past (martingale), so $E(x_t | x_{t-1}, \dots) = x_{t-1} \forall t$.

Extended LIE: $\mathcal{G}, \mathcal{H}$ info sets s.t. $\mathcal{H} \subseteq \mathcal{G}$, then $E(Y | \mathcal{H}) = E(E(Y | \mathcal{G}) | \mathcal{H})$.

If $x_t \in z_t$, $E(x_t | z_{t-1}, \dots) = x_{t-1} \Rightarrow E(x_t | x_{t-1}, \dots) = x_{t-1}$ by extended LIE.

Random Walk: $z_t = z_{t-1} + \varepsilon_t$, $\varepsilon_t \sim iid(0, \sigma^2)$ and z_0 const. $z_t = z_0 + \varepsilon_1 + \dots + \varepsilon_t \Rightarrow E(z_t | \mathcal{F}^{t-1}) = z_{t-1}$. Here, $E(\varepsilon_t | \mathcal{F}^{t-1}) = E(\varepsilon_t | \varepsilon_{t-1}, \dots, z_0) = 0$, bc ε_t iid. $E(z_t) = z_0$, $Var(z_t) = Var(\sum_{i=1}^t \varepsilon_i) = t\sigma^2$.

Martingale diff seq (m.d.s.): $\{z_t\}$ s.t. $E(z_t) = 0$ m.d.s. iff $E(z_t | \mathcal{F}^{t-1}) = 0 \forall t$. Ex: innov of RW, since $E(z_t | \mathcal{F}^{t-1}) = z_{t-1} \Rightarrow E(z_t - z_{t-1} | \mathcal{F}^{t-1}) = 0$. M.d.s. no serial corr, bc $E(z_t z_{t-k}) = E(E(z_t z_{t-k} | z_{t-k})) = E(z_{t-k} E(z_t | z_{t-k})) = 0$ since $E(z_t | z_{t-k}) = E(E(z_t | \mathcal{F}^{t-1}) | z_{t-k}) = 0$ by LIE. So $Cov(z_t, z_{t-k}) = E(z_t z_{t-k}) = 0 \forall k$.

$z_t \sim iid(0, \sigma^2) \Rightarrow z_t$ m.d.s. $E(z_t | \mathcal{F}^{t-1}) = \int z_t f(z_t | \mathcal{F}^{t-1}) dz_t = \int z_t f(z_t) dz_t = E(z_t) = 0$.

$\{z_t\}$ is ARCH(1) if $z_t = \sqrt{h_t} \varepsilon_t$, $\varepsilon_t \sim iid(0, 1)$, $h_t = \alpha_0 + \alpha_1 z_{t-1}^2$, $\alpha_0, \alpha_1 \geq 0$. ARCH(1) is a mds so uncorr, but not indep since $z_t^2 \sim AR(1)$.

ARCH(1) mds: $\mathcal{F}^{t-1} = \{z_{t-1}, \dots, z_1\} = \{\varepsilon_{t-1}, \dots, \varepsilon_2, z_1\}$. Then, $E(z_t | \mathcal{F}^{t-1}) = E(\sqrt{h_t} \varepsilon_t | \mathcal{F}^{t-1}) = \sqrt{h_t} E(\varepsilon_t | \mathcal{F}^{t-1}) = E(\varepsilon_t) = 0$ since ε_t iid and $E(\varepsilon_t) = 0$.

$\exists f$ fn s.t. $z_t = f(z_{t-1}, \varepsilon_t, \dots, \varepsilon_t)$, so $\{z_t\}$ mds $\Rightarrow z_t$ is WN (serially uncorr). $Var(z_t | \mathcal{F}^{t-1}) = E(z_t^2 | \mathcal{F}^{t-1}) = E(h_t \varepsilon_t^2 | \mathcal{F}^{t-1}) = h_t E(\varepsilon_t^2 | \mathcal{F}^{t-1}) = h_t$, since $E(\varepsilon_t^2 | \mathcal{F}^{t-1}) = E(\varepsilon_t^2) = 1$. So h_t is conditional var of z_t , time dependent.

$z_t^2 = \alpha_0 + \alpha_1 z_{t-1}^2 + \eta_t$, where $\eta_t = z_t^2 - h_t = z_t^2 - E(z_t^2 | \mathcal{F}^{t-1})$ is a mds, so WN. $z_t^2 \sim \text{ARCH}(1)$, so ARCH(1) stnry and weak dep. if $|\alpha_1| < 1$ ($\gamma_k = 0, \forall k \neq 0$). Under stnry, $E(E(z_t^2) = E(h_t))$. So $E(h_t) = \alpha_0 + \alpha_1 E(h_{t-1})$, and $\text{Var}(z_t) = E(z_t^2) = E(h_t) = \frac{\alpha_0}{1-\alpha_1}$.

ARCH(p): $z_t = \mu + \sqrt{h_t} \varepsilon_t$, $h_t = \alpha_0 + \alpha_1 z_{t-1}^2 + \dots + \alpha_p z_{t-p}^2$, so $z_t^2 \sim \text{AR}(p)$.

General ARCH mdl of order 1,1 (GARCH(1,1)): Generalise ARCH(1) s.t. $h_t = \alpha_0 + \alpha_1 z_{t-1}^2 + \beta_1 h_{t-1}$, $\alpha_0, \alpha_1, \beta_1 \geq 0$.

$h_t = z_t^2 + (h_t - z_t^2) = z_t^2 - \eta_t$, $\eta_t = z_t^2 - h_t$ is serially uncorr. innov. $z_t^2 = \alpha_0 + (\alpha_1 + \beta_1) z_{t-1}^2 + \eta_t - \beta_1 \eta_{t-1}$, so $z_t^2 \sim \text{ARMA}(1,1)$. z_t indep WN $\Rightarrow z_t$ stnry mds $\Rightarrow z_t$ WN. (G)ARCH not indep WN, since z_t^2 are AR-MA ($\exists$ dep). If z_t mds, $E(z_t | \mathcal{F}^{t-1}) = 0$. If z_t WN, only $\text{Cov}(z_t, z_{t-k}) = 0$, so z_t can be corr w/ non-linear fns of past values.

Unit root procss: $\phi(z) = 1 - z$ has unit root.

RW model ($y_t = y_{t-1} + \varepsilon_t$, $\varepsilon_t \sim iid(0, \sigma^2)$) is a special case. Highly persis (shocks have perm effects), not stnry or weak dep. $\text{Var}(y_t) = t\sigma^2$, so $\text{Corr}(y_t, y_{t+h}) = \sqrt{t/(t+h)}$, if $\text{Var}(y_0) = 0$. Corr dep on t .

$y_{t+h} = y_0 + \sum_{i=1}^{t+h} \varepsilon_i$, so $IRF_h = 1 \forall t, h$. Stn-ry AR(1), $\frac{\partial y_{t+h}}{\partial \varepsilon_t} = \phi_1^h \rightarrow 0$ since $|\phi_1| < 1$.

Forecasting

Optimal MSE forecast of y given x is $E(y|x)$. $\forall g$ fn of x , $MSE(g(x)) = E[(y - g(x))^2] \geq MSE(E(y|x)) = E[(y - E(y|x))^2]$.

$MSE(g(x)) = E[(y - g(x))^2] = E[(y - E(y|x) + E(y|x) - g(x))^2] = E[(y - E(y|x))^2] + 2E[(y - E(y|x))(E(y|x) - g(x))] + E[(E(y|x) - g(x))^2]$. $MSE(E(y|x)) = E[(y - E(y|x))^2]$, $2E[(y - E(y|x))(E(y|x) - g(x))] = 0$ by LIE cond. on x , $E[(E(y|x) - g(x))^2] \geq 0$.

Stnry AR(1): $\hat{y}_{t+1|t} = E(y_{t+1} | \mathcal{F}^t) = E(\phi_1 y_t + \varepsilon_{t+1} | \mathcal{F}^t) = \phi_1 y_t$ (ε_t iid). $h = 1$ forecast: $\hat{y}_{t+1|t} = \phi_1 y_t$. $h = 2$ fore- cast: $\hat{y}_{t+2|t} = E(y_{t+2} | \mathcal{F}^t) = \phi_1 E(y_{t+1} | \mathcal{F}^t) + E(\varepsilon_{t+2} | \mathcal{F}^t) = \phi_1 \hat{y}_{t+1|t} = \phi_1^2 y_t$. So $\forall h \geq 1$, $\hat{y}_{t+h|t} = \phi_1^h y_t$. As $h \rightarrow \infty$, $\hat{y}_{t+h|t} \rightarrow 0 = E(y_{t+h})$ so y_t becomes irrelevant.

RW: $E(y_{t+h} | \mathcal{F}^t) = y_t \forall h \geq 1$, since $y_{t+h} = y_t + \sum_{i=1}^h \varepsilon_{t+i}$ and $E(\varepsilon_{t+j} | \mathcal{F}^t) = 0 \forall j \geq 1$. As $h \rightarrow \infty$, best forecast of y_{t+h} given $\mathcal{F}^t$ is y_t , high persistence. Forecast error:

$y_{t+h} - \hat{y}_{t+h|t} = \varepsilon_{t+1} + \dots + \varepsilon_{t+h} \sim \text{MA}(h-1)$. RW w drift: $y_t = \alpha + y_{t-1} + \varepsilon_t$. $y_t = t\alpha + y_0 + \sum_{i=1}^t \varepsilon_i$, and $E(y_t) = \alpha t$ if $E(y_0) = 0$ (linear). $\hat{y}_{t+h|t} = E(y_{t+h} | \mathcal{F}^t) = h\alpha + y_t$. AR(1) w/ drift: $y_{t+1} = \alpha + \phi_1 y_t + \varepsilon_{t+1}$. $\hat{y}_{t+h|t} = \alpha \sum_{i=0}^{h-1} \phi_1^i = \phi_1^h y_t$. As $h \rightarrow \infty$, this converges to $E(y_t) = \frac{\alpha}{1-\phi_1}$.

3 Panel Data

$y_{it} = \beta_0 + \beta_1 x_{it} + \beta_2 z_i + e_{it}$. OVB: $y_{it} = \beta_0 + \beta_1 x_{it} + \beta_2 z_i + e_{it}$, then $\text{plim} \hat{\beta}_1 - \beta_1 = \beta_2 \frac{\text{Cov}(x_{it}, z_{it})}{\text{Var}(x_i)}$. Error $u_{it} = \beta_2 z_i + e_{it}$ s.t. $\text{Cov}(u_{it}, u_{is}) = \beta_2^2 \text{Var}(z_i)$. FD Est: $\Delta y_i = \beta_1 \Delta x_i + \Delta e_i$, and $\hat{\beta}_1 = \frac{\sum_{i=1}^n \Delta x_i \Delta y_i}{\sum_{i=1}^n (\Delta x_i)^2}$, $\text{plim} \hat{\beta} - \beta \rightarrow^P \frac{E(\Delta x_i e_i)}{E((\Delta x_i)^2)}$. For consis, $E(\Delta e_i \Delta x_i) = 0$, given if strict exog. $\text{Var}(\hat{\beta}_1 | X) = \frac{\sigma^2}{\sum_{i=1}^n (\Delta x_i)^2} = \sigma^2 (X^T X)^{-1}$.

$\Delta y_{it} = \beta_1 \Delta x_{1,it} + \dots + \beta_k \Delta x_{k,it} + \Delta e_{it}$. $\Delta y_i = \Delta X_i \beta + \Delta e_i$, $\hat{\beta} = (\Delta x^T \Delta x)^{-1} \Delta x^T \Delta x = \frac{1}{n(T-1)} (\sum_{i=1}^n \sum_{t=2}^T \Delta x_{it} (\Delta x_{it})^T)^{-1} \cdot (\sum_{i=1}^n \sum_{t=2}^T \Delta x_{it} \Delta y_{it})$. W/ Gauss Markov, FD BLUE.

1. Mdl correct 2. Obs random samp across i 3. Regs change over time 4. Strict exog. $\text{Var}(\Delta e) = E(\Delta e \Delta e^T) = \text{diag}(E(\Delta e_i \Delta e_i^T))$. 5. (Homo) $\text{Var}(\Delta e_{it} | X_i) = \sigma^2$, for SE. True if e_{it} iid, mean 0, var const. 6. (No ser corr) $\text{Cov}(\Delta e_{it}, \Delta e_{is} | X_i) = 0$, not necess true if e_{it} indep., true if RW. Problems: Δx_{it} changes little over time, SE large. To get SE, strong assumpt, might need HAC. W/o 6, not BLUE.

FE: $y_{it} - \bar{y}_i = (x_{it} - \bar{x})^T \beta + (e_{it} - \bar{e}_i)$. $\hat{\beta}_{FE} = (\sum_{i=1}^n \sum_{t=1}^T (x_{it} - \bar{x}_i)(x_{it} - \bar{x}_i)^T)^{-1} \sum_{i=1}^n \sum_{t=1}^T (x_{it} - \bar{x}_i)(y_{it} - \bar{y}_i)$. FE is FD when $T = 2$.

$\hat{y} = \bar{X} \beta + \bar{e}$ (stack time, then entity). $P = X(X^T X)^{-1} X^T$ on $S(X)$, $M = I_n - P$ on $S^\perp(X)$. P, M sym, idempotnt, $PM = 0$, $I = P + M$, $PX = X, MX = 0$.

$M_i = I_T - \frac{1^T 1^T}{T}$, de-means. $\hat{\beta}_{FE} = (\sum_{i=1}^n x_i' M_i x_i)^{-1} \sum_{i=1}^n x_i' M_i y_i$. $\hat{\beta}_{FE} = \beta + (\sum_{i=1}^n \bar{x}_i' \bar{x}_i)^{-1} \sum_{i=1}^n \bar{x}_i' \bar{e}_i$. So $E(\hat{\beta}_{FE} | X) = \beta$ and $\text{Var}(\hat{\beta}_{FE} | X) = \sigma_e^2 (\bar{X}' \bar{X})^{-1}$ if $e_i | X_i \sim iid(0, \sigma_e^2 I_T)$, so no serial/cross-sec corr. Here, FE BLUE. $FE \equiv LSDV$, where dummies for $i \geq 2$. FE more efficient when e_{it} no serial corr. FD more efficient when e_{it} random walk. FE consis if $E(x_{is} e_{it}) = 0$.

2-way: $y_{it} = x_{it}' \beta + \alpha_i + f_t + e_{it}$. If f_t not accounted, OVB. Also, $\text{Cov}(f_t + e_{it}, f_t + e_{jt}) = \text{Var}(f_t) \neq 0$, cross-sec dep.

Can include time dummies if T small. $y_{it} = \delta_1 + \delta_2 d_{2t} + \dots + \delta_T d_{Tt} + x_{it}' \beta + \alpha_i + e_{it}$. FE demeaning: $\bar{y}_{it} = \sum_{i=2}^T \delta_i d_{it}' x_{it}' \beta + \bar{e}_{it}$. $\bar{y}_i = \alpha_i + \bar{f} + \bar{x}_i' \beta + \bar{e}_i$, $\bar{y}_t = \bar{\alpha} + \bar{f}_t + \bar{x}_t' \beta + \bar{e}_t$. (Panel mean) $\bar{y} = \bar{\alpha} + \bar{f} + \bar{x}' \beta + \bar{e}$, so DiD: $y_{it} - \bar{y}_i - \bar{y}_t + \bar{y} = (x_{it} - \bar{x}_i - \bar{x}_t + \bar{x})' \beta + (e_{it} - \bar{e}_i - \bar{e}_t + \bar{e})$, equiv to time dummies & FE. If e_{it} serial dep, FE consis if $E(e_{it} | X) = 0$, but not BLUE. Need cluster SE too. Assume cross-sec indep, but allow $\text{Cov}(e_{it}, e_{is}) \neq 0$, then β unbiased, but SE invalid.

$y_{it} = \mu + e_{it}$, so $\hat{\mu} = 1/Tn \sum_{t=1}^T \sum_{i=1}^n y_{it} = 1/n \sum_{i=1}^n \bar{y}_i$. W/ cross-sec indep by CLT, $\sqrt{n}(\hat{\mu} - \mu) \rightarrow^d N(0, V := \text{Var}(\bar{y}_i))$. $\hat{V} = \frac{1}{n} \sum_{i=1}^n (\bar{y}_i - \bar{y})^2 \rightarrow^P V$ by LLN.

$SE_{\hat{\mu}} = \sqrt{\frac{1}{n^2} \sum_{i=1}^n (\bar{y}_i - \bar{y})^2}$. $V = \frac{1}{T^2} \sum_{t=1}^T \sum_{s=1}^T \text{Cov}(y_{it}, y_{is}) = \frac{1}{T^2} (\sum_{i=1}^T \text{Var}(y_{it}) + 2 \sum_{s < t} \text{Cov}(y_{it}, y_{is}))$. $V = \sigma^2$ if e_{it} iid.

$\hat{V} = \widehat{\text{Var}}(\bar{y}_i) = \frac{1}{n} \sum_{i=1}^n (\bar{y}_i - \bar{y})^2 = \frac{1}{T^2} \sum_{t=1}^T \sum_{s=1}^T (\frac{1}{n} \sum_{i=1}^n (y_{it} - \bar{y})(y_{is} - \bar{y}))$, where inside estimates $\text{Cov}(y_{it}, y_{is})$. $\bar{y}_{it} = \bar{x}_{it} \beta + \bar{e}_{it}$, then $\hat{\beta} = (\sum_{i=1}^n \sum_{t=1}^T (\bar{x}_{it})^2)^{-1} \sum_{i=1}^n \sum_{t=1}^T \bar{x}_{it} \bar{y}_{it}$. $\bar{S}_{xx,i} := \frac{1}{T} \sum_{t=1}^T (\bar{x}_{it})^2$, $\bar{S}_{xe,i} := \frac{1}{T} \sum_{t=1}^T \bar{x}_{it} \bar{e}_{it}$, so $\sqrt{n}(\hat{\beta} - \beta) = (\frac{1}{n} \sum_{i=1}^n \bar{S}_{xx,i})^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n \bar{S}_{xe,i}$.

If $E(\bar{x}_{it} e_{it}) = 0$ (strict exog), $E(\bar{S}_{xe,i}) = \frac{1}{T} \sum_{t=1}^T E(\bar{x}_{it} e_{it}) = 0$. W/ cross-sec indep, $V = \text{Var}(\bar{S}_{xe,i}) = E(\bar{S}_{xe,i}^2)$, and $\hat{V}_{cluster} = \frac{1}{n} \sum_{i=1}^n \frac{1}{T^2} \sum_{t=1}^T \sum_{s=1}^T \bar{x}_{it} \bar{x}_{is} \hat{e}_{it} \hat{e}_{is}$. $\widehat{\text{Var}}(\sqrt{n} \hat{\beta}) = (\frac{1}{n} \sum_{i=1}^n \bar{S}_{xx,i})^{-2} \hat{V}_{cluster}$.

4 Generalised Least Squares

$y = X\beta + e$, $E(e|X) = 0$, $\text{Var}(e|X) = \sigma^2 \Omega$, so OLS not BLUE. New BLUE est is $\hat{\beta}_{GLS} = (X^T \Omega^{-1} X)^{-1} X^T \Omega^{-1} y$.

WTS $\text{Var}(\hat{\beta}_{OLS} | X) - \text{Var}(\hat{\beta}_{GLS}) = (X^T X)^{-1} - \sigma^2 (X^T \Omega^{-1} X)^{-1}$ is pos def.

If Ω sym, pos def, $\exists C$ s.t. $\Omega^{-1} = CC'$, so $C'y = C'X\beta + C'e$, and $y^* = X^* \beta + e^*$. This transformed mdl fulfills Gauss Markov: $E(e^* | X^*) = E(C'e | X) = 0$, $\text{Var}(e^* | X^*) = C^* \text{Var}(e | X) C = C^* \sigma^2 \Omega C = \sigma^2 I_n$.

$\hat{\beta}_{GLS}$ min $Q(\beta) = (y - X\beta)' \Omega^{-1} (y - X\beta)$.

Weighted Least Squares

$\Omega = \text{diag}(\sigma_i^2)$, $\sigma_i^2 = \text{Var}(e_i | X)$, so $Q(\beta) = \sum_{i=1}^n (y_i - x_i' \beta)^2 / \sigma_i^2$. SSR of weighted ob- servations. $C = \text{diag}(\sigma_i^{-1})$, and BLUE.

Model $\sigma_i^2 = E(e_i^2 | X_i) = f(x_i, \theta)$. FGLS: Est β by OLS, $\hat{e}_i = y_i - x_i' \hat{\beta}_{OLS}$. Then $\hat{e}_i^2 = x_i' \theta + v_i$. Reg $\hat{e}_i^2$ on x_i to get $\hat{\theta}_{OLS}$, and $x_i' \hat{\theta} = f(x_i, \hat{\theta}) = \hat{\sigma}_i^2$. Then GLS. AR(1), reg $\hat{e}_t$ on $\hat{e}_{t-1}$ to get $\hat{\rho}$, and run GLS. Random Effects: $y_{it} = \alpha_i + x_{it}' \beta + e_{it}$. If $\alpha_i \perp x_{it}$, $y_{it} = x_{it}' \beta + v_{it}$ where $v_{it} = e_{it} + \alpha_i$. Let $e_{it} \sim iid(0, \sigma_e^2)$, $\alpha \sim iid(0, \sigma_\alpha^2)$, $e_{it} \perp \alpha_i$, then $\text{Var}(v_{it}) = \sigma_e^2 + \sigma_\alpha^2$, and $\text{Cov}(v_{it}, v_{is}) = \sigma_\alpha^2$.

$E(v | X) = 0$, from strict exog and $E(\alpha_i | X) = 0$, OLS unbiat but ineff, use GLS with $\hat{\sigma}_e^2 = \frac{1}{n(T-1)-k} \sum_{t=1}^T \sum_{i=1}^n \hat{e}_{it}^2$ and $\sigma_\alpha^2 = \widehat{\text{Var}}(\bar{u}_i) - \frac{1}{T} \hat{\sigma}_e^2$ where $\widehat{\text{Var}}(\bar{u}_i) = \frac{1}{n-k} \sum_{n-k} \sum_{i=1}^n (\bar{y}_i - \bar{x}_i' \hat{\beta}_{btwn})^2$, from $\bar{y}_i = \bar{x}_i' \beta + \alpha_i + \bar{e}_i$ reg.

FWL: $y = X_1 \beta_1 + X_2 \beta_2 + e$, then reg $M_1 y = M_1 X_2 \beta_2 + u$ to get $\hat{\beta}_2$ and $\bar{u}$. Here, $\hat{\beta}_2 = \hat{\beta}_2$ and $\hat{e} = \bar{u}$.

5 Generalised Method of Moments

Endogenous regs, so $E(x_i e_i) \neq 0$. If $q_i = \alpha_0 + \alpha_1 p_i + e_{1i}$ and $q_i = \beta_0 + \beta_1 p_i + e_{2i}$, regs endogenous and $\text{Cov}(p_i, e_i) \neq 0$. $q_i = \theta_0 + \theta_1 p_i + v_i$ where $\text{Cov}(p_i, v_i) = 0$. $\hat{\theta}_1 \rightarrow^P \frac{\text{Cov}(p_i, q_i)}{\text{Var}(p_i)} = \theta_1 = \frac{\beta_1 \sigma^2 + \alpha_1 \sigma^2}{\sigma_1^2 + \sigma_2^2}$.

Instrumental Variables

Let $q_i = \beta_0 + \beta_1 p_i + \beta_2 t_i + \varepsilon_{2i}$, where $B_2 \neq 0$ and $\text{Cov}(t_i, \varepsilon_{2i}) = 0$. We might want to use t_i as instrumental var for p_i .

Require: $\text{Cov}(t_i, e_{1i}) = 0$ (exogeneity) and $\text{Cov}(t_i, p_i) \neq 0$ (relevance). Relv is identifcatn cond. Can test w/ reduced form eqn. $\text{Cov}(t_i, p_i) = -\frac{\hat{\beta}_2}{\hat{\beta}_1 - \alpha_1} \text{Var}(t_i) \neq 0$.

Let $y_i = \beta_1 + \beta_2 x_{2i} + e_i$, where $E(e_i) = 0$, $\text{Cov}(e_i, x_{2i}) = E(x_{2i} e_i) \neq 0$. z_{2i} IV for x_{2i} if not in eqn above, $\text{Cov}(z_{2i}, e_i) = 0$ and $\text{Cov}(z_{2i}, x_{2i}) \neq 0$.

$\text{Cov}(z_{2i}, y_i) = \beta_2 \text{Cov}(z_{2i}, x_{2i}) + \text{Cov}(z_{2i}, e_i)$, so $\beta_2 = \frac{\text{Cov}(z_{2i}, y_i)}{\text{Cov}(z_{2i}, x_{2i})}$, est w/ $\text{var} - \text{var}$.

$\text{plim} \hat{\beta}_{2,IV} = \beta_2$, consistent. $\hat{\beta}_{1,IV} = \hat{\beta}_{2,IV} \bar{x}_2$. since $\beta_1 = E(y_i) - \beta_2 E(x_{2i})$.

Test $H_0 : \text{Cov}(x_{2i}, x_{2i}) = 0$ w $x_{2i} = \pi_1 + \pi_2 z_{2i} + v_i$, $E(v_i) = 0$, $E(z_{2i} v_i) = 0$. Then $H_0 : \pi_2 = \frac{\text{Cov}(z_{2i}, x_{2i})}{\text{Var}(z_{2i})} = 0$, $H_1 : \pi_2 \neq 0$ with $t_{\hat{\pi}_2} = \hat{\pi}_2 / \text{se}(\hat{\pi}_2)$.

$\sqrt{n}(\hat{\beta}_2 - \beta_2) \rightarrow N(0, V_{\hat{\beta}_2})$, where $V_{\hat{\beta}_2} = \frac{\text{Var}((z_{2i} - \mu_{z_2}) e_i)}{(\text{Cov}(z_{2i}, x_{2i}))^2}$. W/ ho- mo, $\text{Var}((z_{2i} - \mu_{z_2}) e_i) = E((z_{2i} - \mu_{z_2})^2 E(e_i^2 | z_{2i})) = \sigma^2 \text{Var}(z_{2i})$, so

$V_{\hat{\beta}_2} = \frac{\sigma^2}{\rho_{z_2, x_2}^2 \text{Var}(x_{2i})} \cdot V_{\hat{\beta}_2, OLS} = \frac{\sigma^2}{\text{Var}(x_{2i})}$, so IV SE larger than OLS. $\hat{\sigma}^2 = \frac{1}{n-2} \sum_{i=1}^n \hat{e}_i^2$. $y_i = x_i' \beta + e_i$, with k_1 exog k_2 endog variables, so $E(x_{1i} e_i) = 0$ but $E(x_{2i} e_i) \neq 0$. Let z_i be l rvs s.t. $E(z_i e_i) = 0$. Then, z_i is vector combining x_{1i} and l_2 vars. If $k_2 = l_2$, just identified. $E(z_i e_i) = E(z_i (y_i - x_i' \beta)) = 0$, so $E(z_i x_i' \beta) = E(z_i y_i)$. If identified, only 1 val that solves this. Need $\text{rank}(E(z_i x_i')) = k$ so $l \geq k$. Not suff for ident. No. instruments $\geq$ no. regressors, so $\det(E(z_i x_i')) \neq 0$ so no multicoll.

$\hat{\beta}_{IV} = (Z' X)^{-1} Z' y$. Only works when $k = l$ so can invert.

2SLS estimator: When $l = k$, same as before. When $l > k$, most efficient. Reduced form eqns: $x_i = \Pi' z_i + v_i$, so $y_i = (z_i' \Pi) \beta + (e_i + v' \beta)$ where $\Pi = E(z_i z_i')^{-1} E(z_i x_i')$.

$\hat{\Pi} = (Z' Z)^{-1} Z' X$, so $\hat{\beta}_{2sls} = (X' Z (Z' Z)^{-1} Z' X)^{-1} X' Z (Z' Z)^{-1} Z' y = (X' P_Z X)^{-1} X' P_Z y$.

Define $\hat{X} = P_Z X$, then this is OLS on $\hat{X}$.

1. Reg X on Z to get $\hat{X} = P_Z X$ 2. Reg y on $\hat{X}$ to get $\hat{\beta}_{2sls}$. 2SLS chooses best lin comb of IVs, most eff under homo.

Eff IV: $\hat{\beta}_{IV} = (\hat{X} X)^{-1} \hat{X}' y = \hat{\beta}_{2sls}$. $\text{Var}(\hat{\beta}_{2sls} | X, Z) = \sigma^2 (\hat{X}' \hat{X})^{-1}$. $\hat{\sigma}^2 = (y - X \hat{\beta})' (y - X \hat{\beta}) / (n - k)$.

Test rank condition: cannot use F test separately, since same IV can explain multiple x_i 's.

Test for endogeneity, $y_i = x_{1i}' \beta_1 + x_{2i}' \beta_2 + e_i$. Test $H_0 : E(x_{2i}, e_i) = 0$. Compare OLS and 2SLS estimators: $H = \sqrt{n}(\hat{\beta}_{2sls} - \hat{\beta}_{OLS})' [\widehat{\text{Var}}(\sqrt{n}(\hat{\beta}_{2sls} - \hat{\beta}_{OLS}))]^{-1} \sqrt{n}(\hat{\beta}_{2sls} - \hat{\beta}_{OLS})$. If H_0 true, OLS and 2SLS should be similar. Otherwise, $\hat{\beta}_{ols} \rightarrow^P \beta_{proj} \neq \beta \leftarrow^P \hat{\beta}_{2sls}$. H asymp $\sim \chi^2$.

1 regressor: est residuals $\hat{v}_{2i} = x_{2i} - z_i' \hat{\pi}$, then $y_i = x_{1i}' \beta_1 + x_{2i}' \beta_2 + \rho \hat{v}_{2i} + r_i$. Test $H_0 : \rho = 0$. If reject, x_2 exogenous.

Test exog instrument, only if overidenti- fied with J test.

Est residuals $\hat{e}_i$ with 2SLS, regress $\hat{e}_i$ on all exog variables and get R^2 . $J = nR^2$.

Under $H_0 : E(z_i e_i) = 0$, $J \rightarrow^d \chi_{l-k}^2$.

Moment eqn mdls: $E(g_i(\beta)) = 0$. When $l = k$, MoM est: $\bar{g}_n(\hat{\beta}_{mm}) = \frac{1}{n} \sum_{i=1}^n g_i(\hat{\beta}_{mm}) = 0$ matches IV est.

GMM criterion: $J(\beta) = n \bar{g}_n(\beta)' W \bar{g}_n(\beta)$, W pos def weights. Has quad form.

If $g_i(\beta) = z_i(y_i - x_i' \beta)$, $\hat{\beta}_{gmm} = (X' Z W Z' X)^{-1} X' Z W Z' y$.